

# Phase 2 - insight report

Data Vortex Round 1 Phase 2 · what the 21 challenges show about the Social Engine corpus, and which of the obvious readings the data refuses to support.

|           |                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------|
| Team      | Forge-X                                                                                            |
| Team head | Kunal Choudhary                                                                                    |
| Scope     | E1-E5 (easy) · M1-M5 (medium) · H1-H6 (hard)                                                       |
| Data      | 10,221 analysis posts, 1,500 users, 12,000 distinct intake posts                                   |
| Method    | SQL over output/social_engine.db; 21 challenges + 17 verification queries; 43 automated invariants |

## Every answer at a glance

| #  | Challenge                             | Answer                       |
|----|---------------------------------------|------------------------------|
| E1 | Highest-volume platform               | YouTube - 1,770 posts        |
| E2 | Top 10 by likes + shares + comments   | 7,893 at #1                  |
| E3 | Highest average total engagement      | Instagram - 3,673.2 per post |
| E4 | Shares > 1,500, likes < 500           | 214 posts                    |
| E5 | More than 40,000 followers            | 293 users                    |
| M1 | Total engagement by location          | Los Angeles, USA - 1,468,866 |
| M2 | High vs low follower cohorts          | no advantage either way      |
| M3 | Most active users                     | 17 posts for the leader      |
| M4 | Best platform for >= 30k accounts     | Instagram - 3,692.1 per post |
| M5 | Shares > likes + comments             | 1,047 posts                  |
| H1 | Users above 2x the average engagement | no user qualifies            |
| H2 | Top 3 users per location              | 99 rows (3 x 33 locations)   |
| H3 | Posts at 2x their platform mean       | 56 posts                     |
| H4 | Small accounts in the top decile      | 17 users                     |
| H5 | Corrupted posts                       | 4,418 posts                  |
| H6 | Suspicious high-impact users          | 82 users                     |

## Findings

- 1. The platform field carries no preference.** YouTube leads the labelled platforms with 1,770 posts, but against a uniform expectation of 1,736.0 the chi-square statistic is 2.6094 on 4 degrees of freedom - far inside the 9.488 critical value. The five platforms are indistinguishable, so 'YouTube dominates' is a ranking artefact.
- 2. Engagement does not vary meaningfully by platform either.** Instagram leads on average total engagement at 3,673.2 against 3,636.1 for Facebook - a 1.02% gap with  $z = 0.635$ . The ordering is inside the noise band.
- 3. Follower count does not buy engagement.** The high-follower cohort ( $\geq 25,000$ ) averages 3,598.82 per post against 3,647.98 for the low cohort - the smaller accounts are marginally ahead,  $z = -1.441$ . This independently reproduces the recovery phase's correlation of -0.0433 between followers and engagement, by a different route.
- 4. The location league table measures post volume, not audience quality.** Los Angeles, USA tops total engagement with 1,468,866 across 391 posts, but re-ranking by engagement per post moves some locations by up to 27 places. Total engagement is volume x quality, and only one of those is interesting.
- 5. Over-sharing is a real, measurable pattern across the whole corpus.** 214 posts have more than 1,500 shares but fewer than 500 likes, and 1,047 have more shares than likes and comments combined. It is densest on Instagram (2.61% of its posts) but present on every platform, so it is a property of the content, not of one network.
- 6. There is no 2x power-user tier in this corpus.** The highest per-post average any user reaches is 6,545.50 against a 2x bar of 7,247.43 - only 1.81x the baseline. The threshold curve shows where the question does bite: 145 users clear 1.25x, 8 clear 1.5x and 1 clears 1.75x.
- 7. Small accounts are over-represented at the top of the engagement distribution.** 17 users with fewer than 5,000 followers sit in the top decile of total engagement, above a decile floor of 38,765. They also clear the flagship test of high output per follower rather than high reach, which is why the follower-engagement story collapses in both directions.
- 8. The corruption is systematic, fully inventoried, and separable.** 4,418 distinct posts carry at least one defect: 1,770 negative like counts, 1,846 missing platform labels, 1,004 blank text bodies and 525 rows carrying HTML markup at intake. 20 posts carry three defects at once. Every count was recomputed in SQL and matches the recovery phase's own corruption log, so the defects are characterised rather than merely counted.
- 9. A small cohort matches the profile of unusual high-impact accounts.** 82 users combine all three conditions - fewer than 10,000 followers, above-average engagement per post, and at least one post with more shares than likes. They emerge from a funnel of 1,500 users, 287 with a small following and 1,045 owning an over-shared post, so the three filters are each doing work.

## Findings we withdrew

Six plausible results were produced by the queries and then rejected by their own verification queries. They are reported here because the rejections are the analysis: each one is a sentence that would have read as insight.

- **'YouTube dominates the platform mix.'** Withdrawn: chi-square = 2.6094 on 4 degrees of freedom against a 9.488 critical value. The volumes are consistent with no platform preference.
- **'Instagram generates the most engagement.'** Withdrawn as a finding: the mean lead is real but carries  $z = 0.635$ .
- **'Big accounts get more engagement.'** Withdrawn, and the sign is backwards:  $z = -1.441$ .
- **'These are the most engaged locations.'** Withdrawn: the ordering is a post-count ordering, reproducing itself as a ranking shift of up to 27 places.
- **'2,201 posts show abnormal sharing.'** Withdrawn: that number comes from reading a missing like count as zero. 1,154 of those rows - 52.43% of the total - exist only because the field is blank.
- **'Twice the average engagement identifies our power users.'** Withdrawn: the threshold is never reached, so the query returns no rows and the report says so instead of lowering the bar to produce a list.

## Caveats that change how these answers should be used

### 1,532 posts (15.0%) have no like count

They are excluded wherever a challenge demands a like count and retained everywhere else through the NULL-safe engagement expression. Where a result depends on that choice - M5 is the sharpest case - the verification query reports both numbers so the reader can see the sensitivity.

### 1,541 posts (15.08%) have no platform label

They are counted in volume questions because the posts are real, and excluded from every per-platform rate because the label is not. The verification queries price this exclusion: H3b reports the five posts it removes.

### Every threshold in this set is a choice, and each was swept

The follower cuts (25k, 30k, 40k) and the engagement multiplier (2x) were supplied by the challenges. M4b repeats its answer at five follower thresholds and H1c at four multipliers, so a reader can see whether a finding is a property of the data or of the cut.

### What this data cannot support

Engagement here is a count of likes, shares and comments - there are no impressions, no reach and no time series in the challenge questions, so none of these findings is a statement about rate or causality. The corpus is also close to uniform by construction, which is why effect sizes are small and why the honest conclusion is repeatedly 'no detectable difference'.